Operating LC-MARE
(Low Cost Marine Recorder)

05-11-2026

The following notes summarize operating aspects for the LC-MARE recorder.

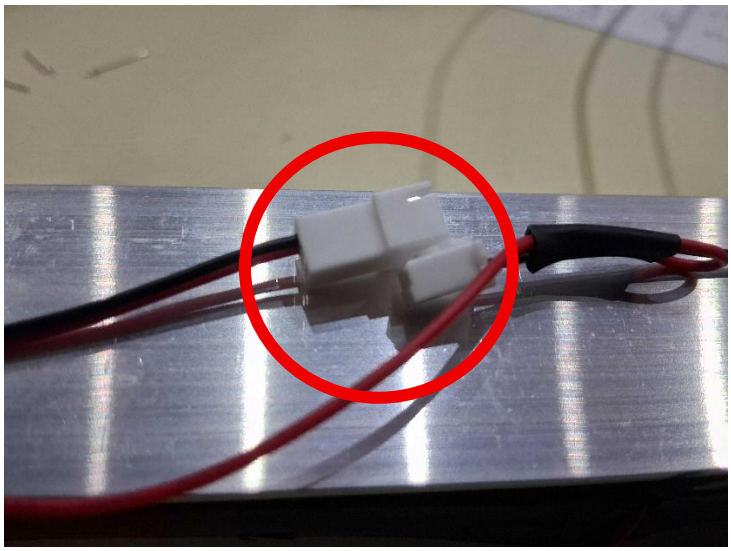


Figure 1Battery pack connector

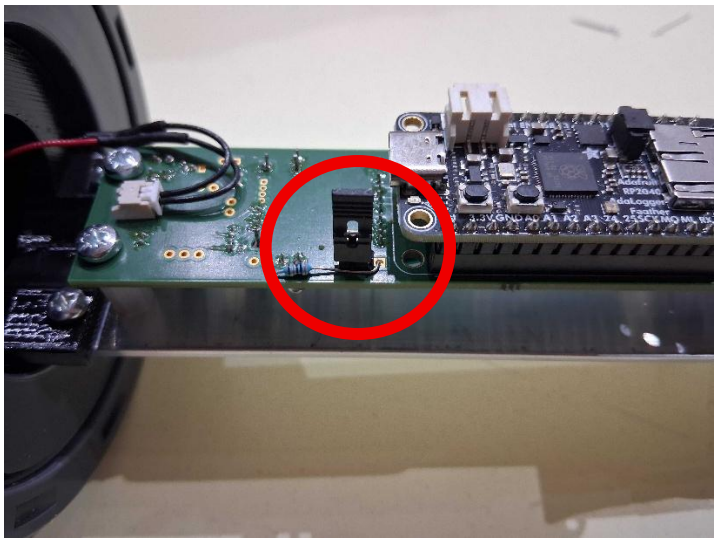


Figure 2: Power supply disable jumper

Before inserting batteries

- Disconnect battery cable connector at the back, as in Fig1.
- Add the two-pin jumper to disable power supply, as in Fig2
- Batteries 6 D-Cell Alkaline in 2 battery holders of 3 (30 to 35 days
  Using only 3 cells in 1 battery holder halves the endurance (15-17 days)

With all 6 batteries recorder is slightly negative buoyant. With 3 batteries recorder should be positive buoyant.

Before connecting to PC

- Ensure battery cable connector at the back is disconnected.
- Remove microSD card (to avoid unnecessary acquisition)
- Connect PC
- Remove jumper to enable power supply (and keep jumper save)
- Ready to use workbench

Using workbench

Typical sequence of operation (more details in Workbench.pdf)

- Connect to device
- In device tab check RTC time
- Sync with PC if necessary
- If necessary, in firmware tab find and upload firmware
- Insert micro-SD card
- In device tab reboot
- After reboot, reconnect to device
- In device press stop acquisition
- In configuration tab
    - Set sampling frequency
    - Set duty cycle
    - Set meta data (organization, operator, project, location)
    - Save to board

Prepare for operation

- Disconnect PC
- Add jumper to disable power supply to MCU
- Connect battery packs
- Insert microSD card (if config.txt is prepared in PC)
- Remove jumper before closing container

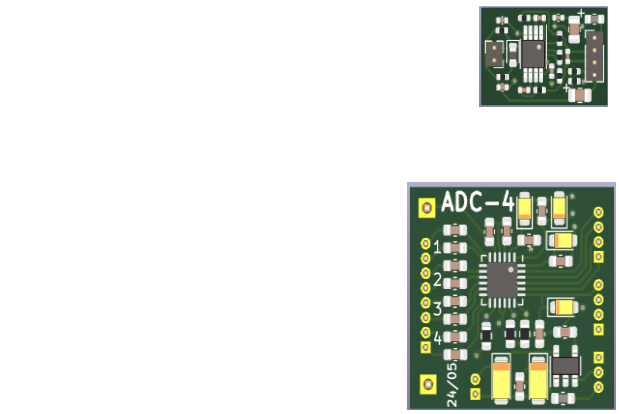
Metadata

keep ISRC as is (it contains hydrophone and sensitivity
NOTE: keep text to less than 39 characters

ICMS: commissioning organisation
IART: operator setting up the recorder (principal investigator)
IPRD: project of PAM operation
ISBJ: area of PAM operation
INAM: location detail of PAM recorder

Electronics

The following shows electronic components used for LC-MARE



Pre-amp
Gain 21
AD8656
25 mW

ADC
4-chan
Up to 768 kHz
TLV320ADC6140
50 mW

Recorder
MCU: RP2040
uSD storage
RV3028 RTC
80 mW @ fs=96 kHz

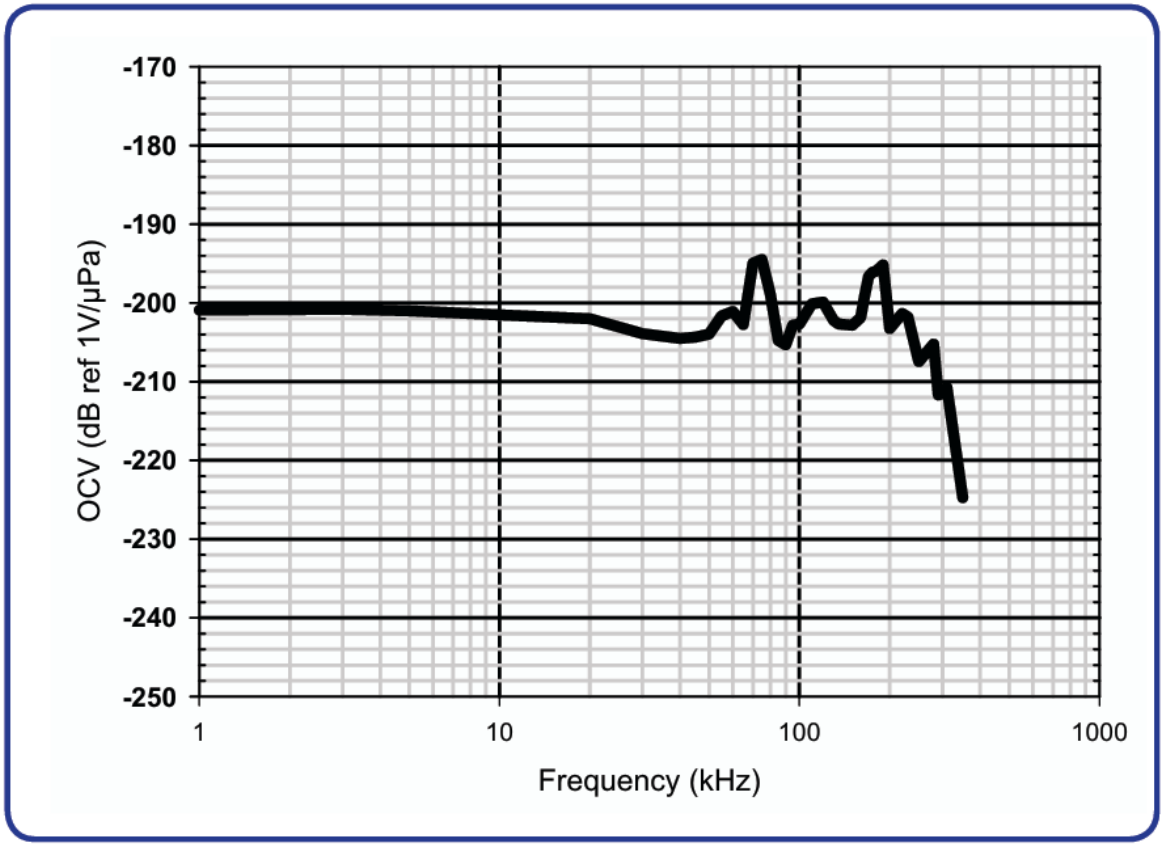
Power+Container
144 Wh Battery
930 h Endurance @ 155 mW
16 bar max pressure

Comments:

microSD card holder is if push-push type (push to insert, push to eject)
RTC battery is of 377 type (or LR66/SR66) may last for years.

SQ17 receiving sensitivity

Following is factory calibration (of SQ17)



Receive Voltage Response